

Notes on ergodicity

1 Introduction: the concept of ergodicity

Does observing a single instance of a system for a long enough time allow one to understand the statistics of a population of such systems? We say a system is ergodic when the answer to this question is yes. Ergodicity justifies generalising temporal observations of an individual experiment, agent or organism to population level statistical statements. It also justifies specialising population level statistical properties to statements about individual members of a population. The scientific study of ergodicity grew out of physics where it provided justification for the use of ensembles in statistical mechanics and out of statistics where it provided justification for the efficacy of statistical inference. The reason researchers needed a word for this property is that it was quickly understood by physicists, statisticians and mathematicians that it does not always hold and should therefore not be treated as an unexamined assumption.

Despite the fundamental and fairly intuitive nature of the concept, it turns out to be difficult to make the notion of ergodicity mathematically precise in a way that applies to all the use cases. This note is my own attempt to piece together the various facets of ergodicity

2 Mathematical formulation of ergodicity for deterministic dynamical systems

Ergodicity is a property of a measure-preserving dynamical system (Ω, μ, T) where

- Ω is the phase space - the set of all possible states the system can be in.
- μ is a measure which assigns a non-negative volume to any subset of the phase space
- $T : \Omega \mapsto \Omega$ is a map which generates dynamics in Ω by iteration:

$$x_{n+1} = T(x_n). \quad (1)$$

Measure-preserving means that

$$\mu(T^{-1}A) = \mu(A) \quad (2)$$

for all subsets $A \subset \Omega$. In other words, the map T does not shrink or enlarge the volume of any set, it just moves sets around. Here $T^{-1}(A)$ denotes the pre-image of A under the map T :

$$T^{-1}(A) = \{x \in \Omega : T(x) \in A\}. \quad (3)$$

In other words, $T^{-1}(A)$ is the set of all points in Ω that are mapped to A by T . A set A is said to be *invariant* if $T^{-1}(A) = A$. Invariant sets are those that remain unchanged under the dynamics T . The use of the pre-image in these definitions is technically important because T need

not be invertible. However for the purposes of building intuition, one can restrict to invertible maps and think of an invariant measure as satisfying $\mu(T(A)) = \mu(A)$ in Eq. (2). In addition to being invariant under T , the measure, μ , is also required to be finite in the sense that $\mu(\Omega) < \infty$. Typically we normalise so that $\mu(\Omega) = 1$. In many examples, the phase space is compact which makes it easier for the total measure to be finite but this is not a requirement.

The definition of ergodicity which has proven most fruitful mathematically is surprisingly abstract: a measure preserving dynamical system is ergodic if the only invariant sets have measure zero or 1.

It is not immediately obvious that this definition is equivalent to the more common (for physicists) definition of ergodicity in terms of the equality of time averages and phase space averages:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x_0) = \int_{\Omega} f(x) d\mu(x) \quad (4)$$

for all integrable functions f on Ω and almost all x_0 . “Almost all” means the set of exceptions to the statement have measure zero. The almost all qualification arises very frequently in precisely stated mathematical results because there are often exceptional cases that do not satisfy the stated property. Asserting that the measure of all such exceptions is zero formalises the idea that the exceptions are atypical and by implication can be ignored for practical purposes.

Physicists often refer to Eq. (4) as Birkhoff's Theorem. However, for mathematicians, the deep part of Birkhoff's contribution is actually a pointwise statement. Birkhoff's (Pointwise) Ergodic Theorem states that for a measure preserving dynamical system, (Ω, μ, T) , the time average of any (integrable) function, f , converges to an invariant function, f^* :

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x_0) = f^*(x_0) \quad (5)$$

for almost all x_0 where $f^* \circ T = f^*$. If, in addition to being measure-preserving, the system (Ω, μ, T) is also ergodic, then it can be further shown that any invariant function must be constant on Ω and, in the case of Eq. (5), that constant must be equal to the phase space average. From a mathematical perspective, the equality of time averages and phase space averages in Eq. (4) is therefore a corollary for ergodic systems of a deeper theorem guaranteeing that time averages converge in the first place. To avoid confusion around exactly what should be called Birkhoff's Theorem, we shall refer to Eq. (4) as simply “Equality of Averages”. Our purpose in mentioning this here is to indicate that the Equality of Averages is not an obvious consequence of the formal

definition of ergodicity in terms of measures of invariant sets. For those interested in a proper account of the mathematical details, see [22].

Ergodicity can also be thought of as a statement about recurrences (how often a trajectory revisits a set $A \subset \Omega$): if you follow a single trajectory of the dynamical system and count how often it is in a set A , the proportion of iterations the trajectory is in A will converge to $\mu(A)$. Mathematically, this follows from applying Equality of Averages, Eq. (4), to the indicator function on A , $1_A(x)$:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} 1_A(T^n x_0) = \int_{\Omega} 1_A(x) d\mu = \mu(A). \quad (6)$$

Recurrence frequencies become stable as N grows and are consistent with the invariant measure. Let us denote by τ_A the random variable corresponding to the time that a trajectory takes to revisit A having left it. This is called the return time. With some technical qualifications, it can be shown [9] that the mean return time, $\mathbb{E}[\tau_A] = \frac{1}{\mu(A)}$. Another way to state Eq. (6) is that for almost all sets, A , the number of recurrent visits to A asymptotically grows proportional to N :

$$\sum_{n=0}^{N-1} 1_A(T^n x_0) \sim cN \quad (7)$$

where $c > 0$ is the inverse of the mean return time.

Ergodicity provides a precise notion of typicality: in an ergodic system almost all initial conditions are typical in the sense that they will eventually explore the full phase space. It is worth emphasising that in this mathematical setting, ergodicity is a property of the map, T , and the invariant measure, μ . For a given map, T , there need not be a unique invariant measure. For example, consider a map which has 2 fixed points, x_1 and x_2 with $T(x_1) = x_1$ and $T(x_2) = x_2$. The point measures $\mu_1 = \delta(x - x_1)$ and $\mu_2 = \delta(x - x_2)$ are both invariant measures with respect to T and both are ergodic. Physically, different invariant measures therefore correspond to different notions of typicality. The fixed points example is trivial here, although in more complex situations, invariant sets can be less trivial than simple fixed points. The measures μ_1 and μ_2 describe the typical behaviour of trajectories that start at one fixed point or the other. In describing this typical behaviour we must recognise that the fixed points themselves may not be typical with respect to another measure on the phase space. This has practical implications for experiments or simulations where the choice of invariant measure may correspond to particular explicit (or implicit) choices that are made in preparing samples or initial conditions. What is then typical behaviour for one experimental protocol might be highly atypical for another. Hence the mathematical question of uniqueness of

the invariant measure and the choice of measure when it is not unique can be more practically consequential than it might first appear.

If a map admits multiple invariant measures, all, some or none of them may be ergodic. However, if a map admits a unique invariant measure, it must be ergodic. This is easy to see by contradiction: assume that a map admits a unique invariant measure, μ and it is *not* ergodic. We continue to assume that T is invertible so we can dispense with the distinction between pre-images and forward-images. If μ is not ergodic, there must exist an invariant set, A , such that $T(A) = A$ and $0 < \mu(A) < 1$. We can therefore define a new measure μ_A which is defined as follows for any set B :

$$\mu_A(B) = \mu(A \cap B). \quad (8)$$

Now for any set, B , we can reason as follows, using the facts that A is an invariant set and μ is an invariant measure:

$$\begin{aligned} \mu_A(T(B)) &= \mu(A \cap T(B)) \\ &= \mu(T(A) \cap T(B)) \\ &= \mu(T(A \cap B)) \\ &= \mu(A \cap B) \\ &= \mu_A(B). \end{aligned}$$

Thus μ_A is an invariant measure which contradicts the assumption that μ is the unique invariant measure. Therefore unique invariant measures must be ergodic. As seen above, the reverse need not be true: ergodic invariant measures need not be unique.

Much of the motivation for studying ergodicity in dynamical systems was motivated by Boltzmann's work on statistical mechanics. However, proving strict ergodicity for the equations of motion governing many-body physical systems turned out to be extremely difficult. This, together with arguments by Lebowitz [10], Goldstein [8] and others, has provoked debate in physics as to whether strict ergodicity is truly required as an additional hypothesis for statistical mechanics to work. For example, in [11], it is shown that important features of equilibrium statistical mechanics are found even for some integrable systems like the Toda model. This is despite the fact that such systems cannot exhibit the molecular chaos usually associated with the ergodic hypothesis. Many physicists now believe that what really matters is that, for a broad class of stationary many-body physical systems, *typical* trajectories correspond to thermodynamic equilibrium states. This viewpoint is sometimes referred to as the typicality account of ergodicity. It is related to, but conceptually distinct from, the notion of typicality introduced above. It does not require atypical trajectories to have measure zero. Rather, it requires that, as the number of degrees of freedom grows, the phase space volume of the set of trajectories that do not correspond to equilibrium states becomes arbitrarily

small as a proportion of the total phase space volume. The typicality account is discussed further in [15].

3 Stationary stochastic processes as dynamical systems

As a physicist, one of the more difficult aspects of ergodic theory is understanding how the above notion of ergodicity developed for deterministic dynamical systems applies to stochastic processes where we again tend to think of ergodicity in terms of the equality of time averages and ensemble averages. In the case of deterministic dynamics, randomness can be introduced only via random choice of initial conditions. We can then think of the measure μ (suitably normalised) as a probability measure on the phase space and we have a random process albeit with the sole source of randomness being the choice of initial condition. Stochastic processes on the other hand have intrinsically random dynamics for example encoded by the transition kernel of a Markov chain or by the explicit presence of a random driving term as found in stochastic differential equations. The definition of ergodicity in terms of invariant sets discussed in the previous section initially doesn't seem natural since the image of any set is random. The key to describing stochastic and deterministic dynamics within a single conceptual framework is the realisation that a very broad class of stochastic processes are equivalent to deterministic dynamical systems on an enlarged phase space. This construction is not unique. A common feature, however, is that the enlarged phase space is always much larger because points in the enlarged phase space correspond to paths or sequences. We now describe two different perspectives

3.1 Canonical (path space) representation of stationary stochastic processes

Consider a stationary, discrete-time stochastic process, $\{X_n\}_{n \in \mathbb{Z}}$, with state space S . Let us denote the stationary distribution of this process by π_* . Introduce the *path space* as the space of bi-infinite sequences of states:

$$\Omega = S^{\mathbb{Z}}. \quad (9)$$

A point ω in the path space corresponds to a trajectory of the original process:

$$\omega = (\dots x_{-2}, x_{-1}, x_0, x_1, x_2 \dots). \quad (10)$$

Any process can be represented as a deterministic dynamical system on the path space where the dynamics are generated by the so-called *shift map*, $\theta : \Omega \rightarrow \Omega$. The shift is defined as follows. For any $\omega \in \Omega$, the n^{th} entry of $\theta \omega$ is

$$(\theta \omega)_n = \omega_{n+1}. \quad (11)$$

The shift map operates like a ticker tape. It advances each path in Ω by a single time step.

To complete the construction of a deterministic dynamical system in the spirit of section 2, we need an invariant measure on the path space Ω . The joint law of the original stochastic process induces a probability measure, μ , on the path space, Ω - essentially assigning a probability to each path based on how likely it is. Since the process is stationary, μ is invariant under the shift operator, θ . The notion of ergodicity introduced in section 2 for measure preserving dynamical systems now applies directly to the shift map, θ , acting on path space, Ω with the induced invariant measure, μ .

The question of uniqueness can be subtle. Given a stationary process, the corresponding induced invariant measure is unique in the sense that the joint law of the stationary process is essentially equivalent to the induced path space measure. However, given an underlying stochastic dynamic, for example a Markov process, it may have more than one stationary distribution. When discussing uniqueness, it is therefore important to specify whether primacy is given to the dynamic or to a specific stationary law. To complicate matters further, even when the stationary law and the corresponding induced invariant measure are unique, the argument of the previous section does not guarantee ergodicity since it is possible for the shift map on path space to admit other invariant measures that have nothing to do with the stationary law.

Leaving this issue aside, if the shifted system is ergodic then Equality of Averages holds:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} F(T^n \omega) = \int_{\Omega} f(\omega) d\mu(\omega) \quad (12)$$

for any trajectory $\omega \in \Omega$ and any integrable function, F on path space. However, when stochastic processes are used in applications or modelling, we do not usually think of time and ensemble averages over trajectories in path space. To obtain the more usual expression of Equality of Averages for stochastic processes, we choose particular observables that pick out regular functions of single variables, $F(\omega) = f((\omega)_0)$. Since this observable depends only on a single coordinate, in this case $(\omega)_0 = x_0$, the integral on the RHS only depends on the one-dimensional marginal of the path space induced measure μ . Moreover, since the process is stationary, this one-dimensional marginal distribution is the stationary distribution of the original stochastic process. We then have

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x_0) = \int_S f(x) d\pi_*(x) = \mathbb{E}_{\pi_*}[f]. \quad (13)$$

This is the more usual form of Equality of Averages as stated for ergodic stochastic processes.

While the canonical representation is general for any stationary process, the induced measure, μ , can be doing

a lot of work. If the process is very simple and all x_i are independent then this induced measure is the product measure $\mu = \pi_*^{\mathbb{Z}}$. For non-trivial cases, however, the measure μ is potentially very complicated indeed since it encodes all dynamical structure, all temporal correlations and all finite-dimensional marginal distributions.

3.2 Skew-product representation of random dynamical systems

The skew-product representation is applicable when a stochastic process can be separated into a deterministic part and a random driving term. The general form is

$$X_{n+1} = F(X_n, \xi_{n+1}), \quad (14)$$

where the X_n take values in a state space, S , and the ξ_n are random variables taking values in a “noise” space Ξ . A very simple example is the discrete Ornstein Uhlenbeck process:

$$x_{n+1} = a x_n + \xi_{n+1}, \quad (15)$$

where the ξ_n are i.i.d. random variables and we assume $|a| < 1$ to ensure the existence of a stationary distribution. To represent stochastic processes of the form Eq. (14) as a deterministic system on a larger phase space, we again use the notion of paths. This time however, a path $\xi = (\dots \xi_{-2}, \xi_{-1}, \xi_0, \xi_1, \xi_2 \dots)$ is a noise realisation. Noise realisations are bi-infinite sequences living in the space $\Xi^{\mathbb{Z}}$. The deterministic dynamical system equivalent to the process Eq. (14) is the skew-product map, Θ , which maps the extended phase space $\Omega = S \times \Xi^{\mathbb{Z}}$ to itself. For a state $x \in S$ and a noise realisation, $\xi \in \Xi^{\mathbb{Z}}$, the skew-product map is defined

$$\Theta(x, \xi) = (\theta \xi, F(x, \xi_1)), \quad (16)$$

where θ is again the shift map, (11), now acting on noise realisations rather than on the trajectories themselves. In the skew-product formulation, the extended state space $\Omega = S \times \Xi^{\mathbb{Z}}$ cleanly separates the dynamical state from the noise. This is advantageous if the driving noise is i.i.d. as is common in many applications. The induced invariant measure on the $\Xi^{\mathbb{Z}}$ component of the extended phase space is then simply the product measure. Any residual complexity of the induced invariant measure is restricted to the state space component, S .

In addition, the separation into state and noise allows one to study what happens to the deterministic part of the dynamics for a fixed realisation of the noise. For a fixed $\xi \in \Xi^{\mathbb{Z}}$, the deterministic evolution on S is given by the n -step evolution map $\varphi(n, \xi) : S \rightarrow S$, often referred to as the cocycle map. This is defined for each point $x \in S$ as

$$\varphi(n, \xi)(x) = F_{\xi_n} \circ \dots \circ F_{\xi_1}(x) \quad (17)$$

where F_{x_i} is shorthand for the function $F(\cdot, \xi_i)$ in Eq. (14). It is here that the sub-field of random dynamical systems differs in perspective from the general study of stochastic processes. By fixing the noise realisation, time-asymptotic dynamical properties of the deterministic evolution map, $\varphi(n, \xi)$, such as invariant sets, attractors and sensitive dependence on initial conditions can be identified and their average properties then studied by averaging over noise realisations later.

The skew-product representation is natural for modelling many physical systems where noise is often thought of as modelling exogenous fluctuations in environmental conditions. In continuous time, it is a very natural way to think about stochastic differential equations without assuming the semi-Martingale property required for Ito calculus. Furthermore, subject to mild assumptions, every Markov process $\{X_n\}_{n \in \mathbb{Z}}$ (and by extension any finite memory recursive processes like ARMA models) also has a skew-product representation:

$$X_{n+1} = F(X_n, U_{n+1}) \quad (18)$$

where the U_n are i.i.d. uniform random variables on $[0, 1]$. To see this, consider a discrete Markov chain on a state space, S , with transition kernel, $P(x, y)$. Given any state, $x \in S$, $P(x, y)$ is a probability distribution over y . Let $\mathcal{F}_x(y)$ be the corresponding cumulative distribution function (CDF). Like any distribution on $\mathbb{R}$, the distribution $P(x, y)$ it can be generated from a uniform random variable using the inverse of its CDF. That is to say

$$y = \mathcal{F}_x^{-1}(u) \quad (19)$$

has the required distribution, $P(x, y)$, where $u \sim \text{Unif}(0, 1)$. Thus in Eq. (18) we can take $F(x, u) = \mathcal{F}_x^{-1}(u)$ to obtain the skew-product representation. This is sometimes called the random mapping representation or functional representation of the Markov chain.

3.3 An explicit example

In this section we use the discrete Ornstein Uhlenbeck process, Eq. (15) as a pedagogical example to show some explicit calculations for the benefit of readers who find explicit examples useful. In the Gaussian case, $\xi_i \sim N(\mu_\xi, \sigma_\xi^2)$, we can use the fact that sums of Gaussian random variables are Gaussian to guess that the stationary law, $\pi_*^a(x)$, is Gaussian:

$$X_n \sim N(m, v) \quad (20)$$

where the mean, m , and variance, v , are to be determined. Then $aX_n \sim N(am, a^2v)$. Using Eq. (15) and the fact that at stationarity, $X_{n+1} \sim X_n$, we quickly find that

$$m = \frac{\mu_\xi}{1 - a} \quad (21)$$

$$v = \frac{\sigma_\xi^2}{1 - a^2}, \quad (22)$$

and the stationary law is

$$\pi_*^a(x) = \sqrt{\frac{1-a^2}{2\pi\sigma_\xi^2}} \exp \left[-\frac{1-a^2}{2\sigma_\xi^2} \left(x - \frac{\mu_\xi}{1-a} \right)^2 \right]. \quad (23)$$

Time averages can also be calculated explicitly. For example, suppose we wish to calculate the time average of X^2 . Assume for simplicity that $\mu_\xi = 0$. By squaring Eq. (15) and rearranging a little one can write:

$$(1-a^2)X_n^2 = \xi_{n+1}^2 + 2aX_n\xi_{n+1} - (X_{n+1}^2 - X_n^2) \quad (24)$$

Now sum from $n = 0$ to $N-1$ and divide by N to obtain the required time average on the left hand side:

$$\begin{aligned} (1-a^2) \frac{1}{N} \sum_{n=0}^{N-1} X_n^2 &= \frac{1}{N} \sum_{n=0}^{N-1} \xi_{n+1}^2 \\ &\quad + 2a \frac{1}{N} \sum_{n=0}^{N-1} X_n \xi_{n+1} \\ &\quad - \frac{X_N^2 - X_0^2}{N} \end{aligned} \quad (25)$$

As $N \rightarrow \infty$, the first term on the right hand side converges to σ_ξ^2 by the Strong Law of Large Numbers. The second term converges to zero by the Martingale Strong Law of Large Numbers. The third term also converges to zero. Intuitively this is because X_N stays of order one almost surely but a full proof is technical. We then conclude that the time average of X^2 is equal to the variance of the stationary distribution of X as expected:

$$\mathbb{T}[X^2] = \frac{1}{N} \sum_{n=0}^{N-1} X_n^2 = \frac{\sigma_\xi^2}{1-a^2} = \mathbb{E}_{\pi_*^a}[X^2]. \quad (26)$$

Since Gaussian examples, tend to be overemphasised due to their tractability, an interesting non-Gaussian case is symmetric Bernoulli noise

$$\xi_i = \begin{cases} 1 & \text{with probability } \frac{1}{2} \\ -1 & \text{with probability } \frac{1}{2}. \end{cases} \quad (27)$$

By considering the extremal trajectories ($\xi_i = 1$ and $\xi_i = -1$ for all i) it is easy to show by direct calculation that the stationary distribution, $\pi_*^a(x)$, of this process must be compactly supported on $\left[-\frac{1}{1-a}, \frac{1}{1-a}\right]$. Following the reasoning of Eq. (25), the time average of X^2 can again be calculated. The calculation is easier because for Bernoulli noise, $\xi_n^2 = 1$ and X_n^2 is explicitly bounded by $\frac{1}{(1-a^2)^2}$. Repeating the previous calculation we find that

$$\mathbb{T}[X^2] = \frac{1}{1-a^2}. \quad (28)$$

Ensemble averages for the Bernoulli case are difficult to calculate explicitly because the nature of the stationary distribution, $\pi_*^a(x)$, as a function of a is surprisingly complicated. It was characterised in [18] as follows:

- $0 < a < \frac{1}{2}$: $\pi_*^a(x)$ is singular and supported on a Cantor set of dimension $\log 2 / \log a^{-1}$.
- $a = \frac{1}{2}$: $\pi_*^a(x)$ is the uniform Lebesgue measure on $[-2, 2]$
- $\frac{1}{2} < a < 1$: $\pi_*^a(x)$ is absolutely continuous with respect to Lebesgue measure except for a set of values of a of measure zero where it is again singular.

It turns out that for this process, $\pi_*^a(x)$ is unique and the corresponding induced measure on path space is ergodic, even though the measure itself is complicated. The uniform case, $a = \frac{1}{2}$ is the only case for which explicit calculations are simple. For example, when $a = \frac{1}{2}$,

$$\mathbb{E}_{\pi_*^{\frac{1}{2}}}[X^2] = \int_{-2}^2 \frac{1}{4} x^2 dx = \frac{4}{3}. \quad (29)$$

This agrees with the time-averaged variance given by Eq. (28) with $a = \frac{1}{2}$, as it must since the process is ergodic.

4 Infinite ergodicity

Infinite ergodic theory extends some aspects of ergodic theory to recurrent dynamics for which an invariant measure exists but is not normalisable, $\mu(\Omega) = \infty$. A characteristic feature is that almost all sets, $A \subset \Omega$, are visited infinitely often but the mean return time, $\mathbb{E}[\tau_A]$, is infinite. Mathematically, this happens when the distribution of τ_A has sufficiently heavy tails that $\mathbb{E}[\tau_A]$ does not exist. Physically it means that for almost all sets, A , the dynamics visits A infinitely many times but nevertheless spends only a vanishing proportion of the *total* time in A . This is possible because excursions from A can be very long.

Equality of Averages is often trivial because time averages and recurrence frequencies tend to zero. Infinite ergodic theory then has less to do with equality of time and ensemble averages. Instead of the time-average of an observable, f , the central object of study is the time cumulative sum of f :

$$S_N f(x) = \sum_{n=0}^{N-1} f(T^n x). \quad (30)$$

For finite ergodic systems, time cumulative sums grow linearly with N at a rate which is asymptotically deterministic:

$$S_N f(x) \sim N \int_{\Omega} f d\mu. \quad (31)$$

In infinite ergodic systems, by contrast, time cumulative sums grow sub-linearly with N at a rate which remains random even asymptotically:

$$S_N f(x) \sim a_N \zeta \quad (32)$$

where $a_N < N$ and ζ is a random variable with a non-trivial distribution. The sub-linear scaling of time cumulative sums explains why ordinary time averages and recurrence frequencies go to zero in such systems. Asymptotic randomness means that different trajectories retain order 1 differences forever. This feature makes infinite ergodic systems fundamentally different from a physical perspective.

In stochastic processes, the first example of infinite recurrence times that often comes to mind is the simple random walk on $\mathbb{Z}$. Does the simple random walk provide an illustrative example of infinite ergodicity? I would argue this is not immediately obvious because the simple random walk is not stationary. We cannot proceed as in the previous section by taking the invariant measure to be the probability measure induced on path space by the stationary probability distribution of the stochastic process. However we do not require the invariant measure to be a probability measure (normalisable). What alternative does this enable? The transition probabilities for the simple random walk are:

$$P(i, j) = \begin{cases} \frac{1}{2} & \text{if } i = j + 1 \\ \frac{1}{2} & \text{if } i = j - 1 \\ 0 & \text{otherwise.} \end{cases} \quad (33)$$

The so-called counting measure on $\mathbb{Z}$ is defined as follows. For any $A \subset \mathbb{Z}$:

$$\mu(A) = \begin{cases} |A| & \text{if } A \text{ is finite} \\ \infty & \text{otherwise} \end{cases} \quad (34)$$

In particular, $\mu(\{i\}) = 1$ for all $i \in \mathbb{Z}$. Note that although the counting measure on $\mathbb{Z}$ is not normalisable, it is invariant with respect to the transition probabilities Eq. (33):

$$\begin{aligned} \sum_{j \in \mathbb{Z}} P(i, j) \mu(\{j\}) &= \frac{1}{2} \mu(\{i - 1\}) + \frac{1}{2} \mu(\{i + 1\}) \\ &= \frac{1}{2} + \frac{1}{2} = \mu(\{i\}). \end{aligned}$$

Physically, this simply means that at each step, there is as much probability mass entering each site $i \in \mathbb{Z}$ from neighbouring sites as there is leaving due to hopping. The counting measure on $\mathbb{Z}$ can now be used to induce an invariant (but non-normalisable) measure on the path space of the simple random walk. A slightly unintuitive aspect of this construction is that this induced measure on path space is stationary even though the original stochastic process is not. By taking $f(x) = 1_{\{0\}}(x)$ in Eq. (30), we can count the number of visits to site 0 after time N . From the theory of random walks one finds

$$S_N f \sim c \sqrt{N} \quad (35)$$

where c has a non-trivial (half normal?) distribution. Thus ordinary frequencies vanish but normalised occupation counts remain meaningful.

To avoid giving the impression that infinite ergodicity is a complicated way of describing properties of random walks that are obtainable by other means, it is worth briefly considering a second example: the Pomeau-Manneville map. This is a deterministic dynamical system with a compact phase space, $[0, 1]$ where infinite invariant measure arises in a very different way to the simple random walk. The map is

$$T(x) = x + x^{1+\alpha} \mod 1 \quad (36)$$

where $x \in [0, 1]$ and $\alpha > 0$. The important feature is that the fixed point at 0 is neutrally stable because $T'(0) = 1$. Consequently trajectories that wander close to zero can get trapped for very long periods of time. How long depends on the parameter α . One can show that an invariant measure exists and that near zero it behaves like

$$\rho(x) \sim x^{-\alpha}, \quad (37)$$

which reflects the fact that trajectories have a tendency to accumulate in the vicinity of the “sticky” neutral fixed point. If we consider a set which does not include the neutral fixed point at 0, for example $A = [\frac{1}{4}, \frac{3}{4}]$, it is possible to show that the distribution of the return time, τ_A , has power law tails:

$$\mathbb{P}(\tau_A > n) \sim n^{-\frac{1}{\alpha}}. \quad (38)$$

These tails are the result of some trajectories getting trapped near the fixed point and taking a long time to escape. The parameter value $\alpha = 1$ is special. If $\alpha < 1$ the return time distribution is still integrable, the invariant measure is finite and ordinary ergodic theory applies. If $\alpha \geq 1$, the mean return time diverges and we enter the infinite ergodic theory regime. Note from Eq. (37) that the invariant measure becomes infinite since $\int_0^\epsilon \rho(x) dx$ diverges for any $\epsilon > 0$. In this example, infinite measure arises due to the loss of an integrable singularity at a single point in phase space rather than due to trajectories spreading out over infinitely large regions in the case of the random walk example. Nevertheless many of the same phenomena are observed as we noted for the random walk example. Cumulative occupation counts again scale sub-linearly and after appropriate normalisation converge to a nontrivial random limit. This is the aspect that generalises across many examples: infinite ergodic theory is fundamentally about how power law recurrence time statistics lead to persistent randomness in a broad class of deterministic and random dynamical systems.

For further discussion of infinite ergodicity and a more detailed example, see [7]. This paper contains explicit calculations for a class of stochastic process which generalises Geometric Brownian motion to include power law drift and diffusion terms. It is shown how appropriate normalisation can be used to extract physically useful information about the long time asymptotics of such processes even when the corresponding Fokker-Planck equation has no normalisable solution.

5 Nonstationary stochastic processes

Most nonstationary processes are not recurrent, so the notion of infinite ergodicity introduced in Sec. 4 does not apply to them. It is arguably misleading to refer to such nonstationary processes as “non-ergodic” since ergodicity as described in Sections 2 and 3 is a property of stationary systems, and non-recurrence precludes the more general notion of infinite ergodicity. Nevertheless in some applications, authors do emphasise the non-ergodicity of some “obviously non-ergodic” nonstationary dynamics. This can be because the question of whether individual realisations are representative of the ensemble remains important. It can also be because it is possible to find a transformation which isolates a useful ergodic component of the dynamics, either exactly or approximately.

An illustrative example is provided by the multiplicative random walk [14]:

$$X_{n+1} = z_n X_n \quad (39)$$

with $X_0 > 0$ and the z_i being iid non-negative random variables with finite mean and variance (the case $z_i \sim \text{Bernoulli}[\frac{3}{5}, \frac{3}{2}]$ is the multiplicative cointoss example.) We have

$$X_n = X_0 \prod_{i=1}^{n-1} z_i \quad (40)$$

which is clearly nonstationary and therefore nonergodic. Let us take the logarithm and divide by n :

$$\frac{\log X_n - \log X_0}{n} = \frac{1}{n} \sum_{i=1}^{n-1} \log z_i. \quad (41)$$

Taking the logarithm transforms a multiplicative random walk into an additive random walk. If we now take the limit of large n , the law of large numbers gives

$$\lim_{n \rightarrow \infty} \frac{1}{n} (\log X_n - \log X_0) = \mathbb{E}[\log z] \quad (42)$$

almost surely. Therefore, as n gets large, a typical realisation of Eq. (39) almost surely behaves like

$$X_n \sim X_0 e^{\lambda n} \quad (43)$$

where $\lambda = \mathbb{E}[\log z]$ is often referred to as the Lyapunov exponent (since the typical difference between two different initial conditions $X_0^{(1)}$ and $X_0^{(2)}$ grows as $|X_0^{(1)} - X_0^{(2)}| e^{\lambda n}$. On the other hand, since the z_i are iid, the expected value of X_n is given by

$$\mathbb{E}[X_n] = X_0 (\mathbb{E}[z])^n. \quad (44)$$

Therefore as n gets large

$$\mathbb{E}[X_n] \sim X_0 e^{\mu n}, \quad (45)$$

where $\mu = \log \mathbb{E}[z]$. Since the logarithm is convex, Jensen's inequality tells us that $\lambda < \mu$ in general. Typical trajectories are therefore not representative of the ensemble in the sense that the expected value grows exponentially faster with n than individual trajectories. The reason is that the expected value is dominated by large fluctuations which become exponentially rare as n grows but make exponentially large contributions to the average. This mechanism is not a peculiarity of the multiplicative random walk. It appears in many areas of statistical physics such as localisation phenomena, glass dynamics, turbulence and random matrix products. The multiplicative random walk is just the simplest exactly solvable model of a phenomenon that has wide applicability.

Returning to the discussion of ergodicity, the difference between individual trajectories and the ensemble averaged trajectory of the multiplicative random walk is not directly interpretable as a difference of time and ensemble averages. The time average of X_n is either zero or infinite (depending on the sign of λ .) The ensemble average is n dependent and also becomes either zero or infinite as $n \rightarrow \infty$ (depending on the sign of μ .) However, there is an underlying Birkhoff style Equality of Averages ergodicity statement encoded in Eq. (42). To see this, define a new variable,

$$\lambda_n = \log X_{n+1} - \log X_n, \quad (46)$$

corresponding to the increment of $\log X_n$, or equivalently the local in time growth rate of X_n . From Eq. (39) and the fact that the z_i are iid, we note that $\mathbb{E}[\lambda_n]$ is independent of n and given by $\mathbb{E}[\lambda] = \mathbb{E}[\log z]$. The left hand side of Eq. (42) is a telescopic sum: $\log X_n - \log X_0 = \log X_n - \log X_{n-1} + \log X_{n-1} - \dots + \log X_1 - \log X_0$. Putting this together, Eq. (42) tells us that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \lambda_i = \mathbb{E}[\lambda]. \quad (47)$$

This is Equality of Averages, but for the growth rate of X_n rather than for X_n itself. Note what has happened here: although X_n is trivially non-ergodic, by identifying a transformation of X_n which does possess the ergodicity property as per Eq. (47), we are able to understand the temporal behaviour of typical realisations of X_n as per Eq. (43).

Empirical analogues of this theoretical procedure abound in practical time series analysis and signal processing. Transformations are commonly used to attempt to isolate a component of the dynamics that is stationary or ergodic (or approximately so in empirical contexts) for observables of interest. The objective is usually to find transformations which can be used to improve some a-priori measure of predictive power. In applications, detrending and/or seasonable adjustment may be applied to remove systematic drift or periodic oscillation, differencing may be applied to remove

stochastic drift and data transformations may be used to stabilise variance over time. A very commonly used transformation for variance stabilisation is the Box-Cox transformation (which includes the log transformation discussed above as a special case):

$$f_{\eta}(x) = \begin{cases} \frac{x^{1-\eta}-1}{1-\eta} & \text{if } \eta \neq 1 \\ \log x & \text{if } \eta = 1 \end{cases} \quad (48)$$

and its generalisations. Detrending, seasonal adjustment, differencing and data transformation are often combined to find quantities that best achieve this end. This is reflected in the exactly solvable example of the multiplicative random walk where differencing of the log of the signal is required to obtain the quantity, λ_n , that characterises the typical behaviour of individual realisations of the process. For a more complicated example in the present collection, see [16] which exhibits appropriate transformations for processes which combine additive and multiplicative increments. A more empirical approach is described in [3] which discusses how to find appropriate transformations from data.

6 Practical considerations

In rigorous ergodic theory, ergodicity requires answering a question about the global invariant structure of an entire dynamical system (Ω, μ, T) : do all invariant sets have measure 0 or 1? In applications, researchers ask a simpler question: are empirical estimators computed from a single trajectory representative of the ensemble of possible trajectories? As we have discussed, these two questions are closely related theoretically. However, it is often possible to answer the second question approximately without knowing much about the answer to the first one.

In practice, only a finite number of trajectories are observed and each is only of finite length, only a limited number of observables are measured and the true underlying dynamics may not even be known. Under such circumstances it is pretty hopeless to attempt to establish indecomposability of an invariant measure. For this reason, weaker or approximate notions of ergodicity are sometimes used in applied work:

- ergodicity of the mean: does the empirical time average of a quantity of interest equal to its empirical mean?
- ergodicity of higher moments (or of distribution)
- ergodicity as an approximate (FAPP etc) or non-binary property (ergodicity spectrum etc)

These weaker notions of ergodicity provide useful ways to frame important questions in different domains of application. They are distinct, and more operational in nature, than the definition of measure-theoretic ergodicity that has ultimately allowed mathematicians to unify the study of invariant structures in deterministic dynamical systems, stochastic processes and beyond. However this should not preclude

their use and more widespread adoption among applied scientists. The question of when the statistics of individual trajectories are representative of populations and vice versa is a critical one in many fields. The concept of ergodicity too important to be solely an abstraction of mathematics and theoretical physics.

The papers in this collection: [4, 12, 6, 19, 21, 1, 20, 3, 15, 5, 13, 11, 17, 2, 7, 16]

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